

PROJECT PHOENIX

REINFORCED CONCRETE BEAM PRELIMINARY DESIGN REPORT

Beam: RCB-SUR-DEMO-001 - Suriname sketch-input demonstration beam

Project: PHX-SUR-SKETCH-DEMO-001

Status: PRELIMINARY STRUCTURAL DESIGN - ENGINEER REVIEW REQUIRED

Geometry: L=5.0 m, b=300 mm, h=500 mm

Materials: C30/37 / B500B

Standard profile: SUR_STRUCTURAL_PRELIMINARY_2026__PROJECT-SPECIFIED EC2 PRELIMINARY
BASIS FOR DEMONSTRATION

This report is not a signed structural calculation and is not approved for construction.

1. LOADS AND ANALYSIS

Self weight: 3.75 kN/m

Total permanent UDL: 3.75 kN/m

Variable UDL: 12.0 kN/m

ULS reaction A: 80.156 kN

ULS reaction B: 72.656 kN

ULS maximum moment: 114.188 kNm

ULS maximum shear: 80.156 kN

Point loads:

- P-01: 25.0 kN at 2.0 m

2. FLEXURE AND SHEAR

Required steel: 629.563 mm²

Provided steel: 678.584 mm²

Bottom reinforcement: 6T12 bottom ($A_s=679$ mm²; clear=26 mm)

Moment resistance: 123.709 kNm

Flexure utilization: 0.923

Concrete shear resistance: 66.428 kN

Stirrups: 2-leg T8 stirrups @ 200 mm

Governing shear resistance: 87.331 kN

Shear utilization: 0.918

3. SERVICEABILITY AND DETAILING

Estimated deflection: 6.746 mm

Deflection limit: 20.0 mm

Estimated crack width: 0.093 mm

Crack width limit: 0.3 mm

Design anchorage length: 395 mm

Available anchorage length: 850 mm

The serviceability calculations are preliminary estimates and require engineer confirmation.

4. DESIGN CHECKS

CHK-001 | Statics | PASS | Reactions equal total ULS load.

CHK-002 | Flexure resistance | PASS | $M_{Ed}/M_{Rd} = 0.923$

CHK-003 | Neutral axis | PASS | $x/(x_{lim} \cdot d) = 0.333$

CHK-004 | Minimum reinforcement | PASS | $A_{s,prov} = 678.6 \text{ mm}^2$; $A_{s,min} = 200.9 \text{ mm}^2$

CHK-005 | Maximum reinforcement | PASS | $A_{s,prov} = 678.6 \text{ mm}^2$; $A_{s,max} = 6000.0 \text{ mm}^2$

CHK-006 | Bar fit | PASS | 6T12 bottom ($A_s=679 \text{ mm}^2$; clear=26 mm)

CHK-007 | Shear resistance | PASS | $V_{Ed}/V_{Rd} = 0.918$

CHK-008 | Shear compression strut | PASS | $V_{Ed} = 80.2 \text{ kN}$; $V_{Rd,max} = 538.0 \text{ kN}$

CHK-009 | Deflection | PASS | $\delta = 6.75 \text{ mm}$; limit = 20.00 mm

CHK-010 | Crack width | PASS | $w_k = 0.093 \text{ mm}$; limit = 0.300 mm

CHK-011 | Anchorage fit | PASS | $l_{b,design} = 395 \text{ mm}$; available = 850 mm

CHK-012 | Support bearing | PASS | bearing utilization = 0.087

CHK-013 | Deep-section review gate | PASS | $h = 500 \text{ mm}$; mandatory review threshold = 700 mm

CHK-014 | Fire design boundary | PASS | Fire design is excluded and must be completed separately.

CHK-015 | Professional review gate | PASS | All outputs remain preliminary until signed by a competent structural engineer.

CHK-016 | Standard profile declared | PASS |

SUR_STRUCTURAL_PRELIMINARY_2026__PROJECT-SPECIFIED EC2 PRELIMINARY BASIS FOR DEMONSTRATION

5. LIMITATIONS AND NEXT GATE

- This is a preliminary design engine and not a signed structural calculation.
- The selected standard profile and National Annex parameters must be confirmed for the project.
- Fire resistance, robustness, seismic design, fatigue and accidental actions are outside this v1.0.0 scope.
- Construction tolerances, bar curtailment, laps, couplers, support-zone detailing and actual load paths require engineer review.
- The 2025 Dutch National Annex introduced shear-related changes; deep sections trigger a mandatory review gate.

Next gate:

Assign the beam to an approved model object, replace example loads with project loads, and obtain signed structural-engineer review before use for permit, tender or execution.